

VEDAS COLLEGE

AFFILIATED TO TRIBHUVAN UNIVERSITY

JAWALAKHEL, LALITPUR, Nepal



A

Lab Report

On

Simulation and Modelling
(CSC328)

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Faculty: Science and Technology

Level: Bachelor

Program: Computer Science and Information Technology.

Submitted To:

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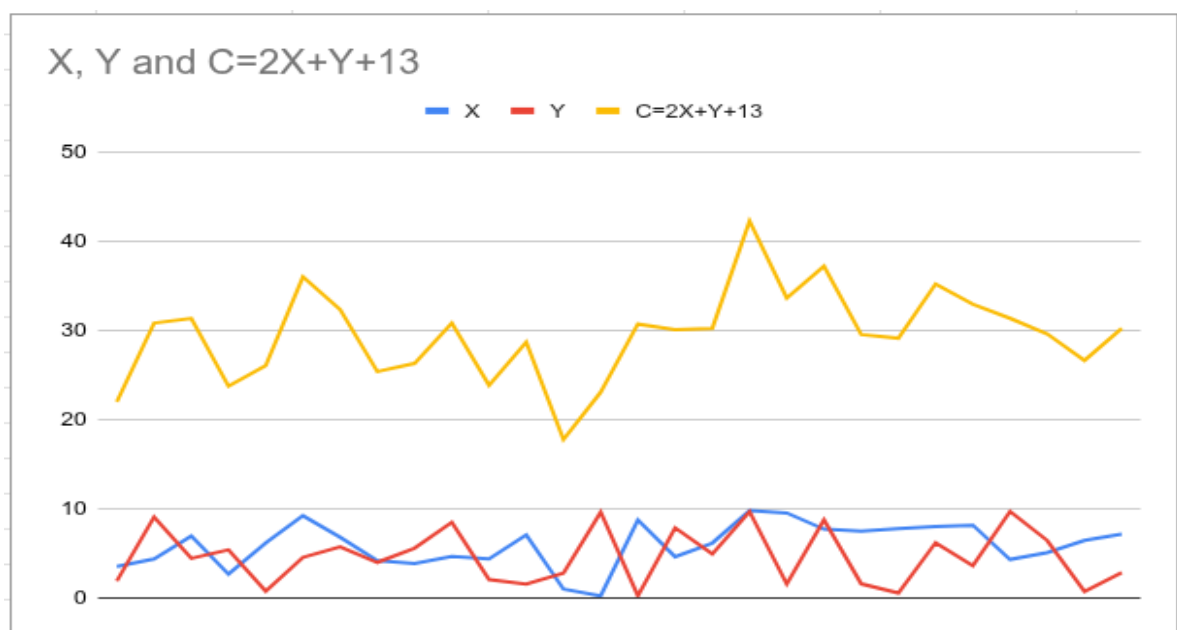
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1. Generate the random numbers using MS-Excel Function.

	A	B	C
1	X	Y	$C=2X+Y+13$
2	3.541934576	1.903570951	21.9874401
3	4.36852987	9.068125893	30.80518563
4	6.939358565	4.447331877	31.32604901
5	2.661366766	5.397527673	23.7202612
6	6.152867807	0.7488577982	26.05459341
7	9.218091247	4.552620005	35.9888025
8	6.789613725	5.725675207	32.30490266
9	4.186552383	3.990229682	25.36333445
10	3.845141878	5.591355715	26.28163947
11	4.657731208	8.477522513	30.79298493
12	4.383356324	2.067554073	23.83426672
13	7.064498643	1.555338026	28.68433531
14	0.9860733643	2.778187854	17.75033458
15	0.2244006425	9.6071163	23.05591758
16	8.730357319	0.2326427638	30.6933574
17	4.604512105	7.858189925	30.06721414
18	6.15404212	4.916331243	30.22441548
19	9.79496088	9.665312157	42.25523392
20	9.527463298	1.556020497	33.61094709
21	7.715346903	8.754222927	37.18491673
22	7.467503485	1.584750505	29.51975747
23	7.778102377	0.5476084405	29.10381319
24	7.994948551	6.19884132	35.18873842
25	8.159372225	3.602108083	32.92085253
26	4.315415225	9.710657683	31.34148813
27	5.076894714	6.427357956	29.58114738
28	6.446588961	0.7199002497	26.61307817
29	7.172616582	2.860037034	30.2052702
30	9.150273349	2.048922405	33.3494691

Output:-



2. Write a C program to find the growth in national consumption for five years using Distributed Lag Model given below:

$$I = 2 + 0.1 Y_1$$

$$Y = 45.45 + 2.27 (I + G)$$

$$T = 0.2 Y$$

$$C = 20 + 0.7 (YT)$$

Assume the initial value of Y_1 is 80 and take the governmental expenditure in the 5 years to be as follows:

Year	G
1	20
2	25
3	30
4	35
5	40

(i). Program:

```
#include <stdio.h>

int main()
{
    int year;
    double Y1 = 80;
    double G[5] = {20, 25, 30, 35, 40};
    double I, Y, T, C;
    printf("Year\tG\tI\tY\tT\tC\n");
    for (year = 0; year < 5; year++)
    {
        I = 2 + 0.1 * Y1;
        Y = 45.45 + 2.27 * (I + G[year]);
        T = 0.2 * Y;
        C = 20 + 0.7 * (Y - T);
        printf("%d\t%.2f\t%.2f\t%.2f\t%.2f\t%.2f\n", year + 1, G[year], I, Y, T, C);
        Y1 = Y;
    }
}
```

```
}  
  
return 0;  
  
}
```

(ii). Output:

```
C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users  
V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 1.c -  
o 1 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab  
2\1"  
Year      G      I      Y      T      C  
1         20.00   10.00   113.55  22.71   83.59  
2         25.00   13.36   132.52  26.50   94.21  
3         30.00   15.25   148.17  29.63   102.98  
4         35.00   16.82   163.07  32.61   111.32  
5         40.00   18.31   177.81  35.56   119.57  
  
c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>
```

3. Write a C program to generate the random numbers using linear congruential generator method.

(i). Program:

```
#include <stdio.h>  
  
int main()  
{  
    long long x0, a, c, m;  
    int n;  
  
    printf("Enter the seed value (x0): ");  
    scanf("%lld", &x0);  
  
    printf("Enter the multiplier (a): ");  
    scanf("%lld", &a);  
  
    printf("Enter the increment (c): ");  
    scanf("%lld", &c);
```

```

printf("Enter the modulus (m): ");
scanf("%lld", &m);

if (m <= 0)
{
    printf("Modulus must be greater than 0\n");
    return 1;
}
printf("How many random numbers to generate? ");
scanf("%d", &n);

printf("Generated random numbers:\n");
for (int i = 0; i < n; i++)
{
    x0 = (a * x0 + c) % m;
    float r = (float)x0 / m;
    printf("%f\n", r);
}
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 2.c -o 2 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\"2
Enter the seed value (x0): 5
Enter the multiplier (a): 15
Enter the increment (c): 8
Enter the modulus (m): 14
How many random numbers to generate? 7
Generated random numbers:
0.928571
0.500000
0.071429
0.642857
0.214286
0.785714
0.357143
c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

4. Write a C program to implement the Kolmogorov-Smirnov Test.

(i). Program:

```
#include <stdio.h>
#include <math.h>

void sort(float x[], int n)
{
    float temp;
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if (x[j] > x[j + 1])
            {
                temp = x[j];
                x[j] = x[j + 1];
                x[j + 1] = temp;
            }
        }
    }
}

int main()
{
    int n;
    float x[100], d_plus = 0, d_minus = 0, d, d_crit;

    printf("Enter number of observations: ");
    scanf("%d", &n);

    printf("Enter the data (values between 0 and 1):\n");
    for (int i = 0; i < n; i++)
```

```

{
    scanf("%f", &x[i]);
}
sort(x, n);

for (int i = 0; i < n; i++)
{
    float d1 = ((float)(i + 1) / n) - x[i];
    float d2 = x[i] - ((float)i / n);

    if (d1 > d_plus)
        d_plus = d1;

    if (d2 > d_minus)
        d_minus = d2;
}
d = (d_plus > d_minus) ? d_plus : d_minus;
d_crit = 1.36 / sqrt(n);
printf("\nD+ = %f", d_plus);
printf("\nD- = %f", d_minus);
printf("\nD = %f", d);
printf("\nCritical Value = %f\n", d_crit);
if (d < d_crit)
    printf("Result: Accept Null Hypothesis (Good fit)\n");
else
    printf("Result: Reject Null Hypothesis (Not a good fit)\n");
return 0;
}

```

(ii). Output:

```
C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users
\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 3.c -
o 3 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab
2\"3
Enter number of observations: 5
Enter the data (values between 0 and 1):
0.10
0.50
0.60
0.70
0.80

D+ = 0.200000
D- = 0.300000
D = 0.300000
Critical Value = 0.608211
Result: Accept Null Hypothesis (Good fit)

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>
```

5. Write a C program to implement the Chi-Square Test.

(i). Program:

```
#include <stdio.h>
#include <math.h>

int main()
{
    int n, i;
    double observed[100], expected[100], chi_square = 0.0;
    printf("Enter the number of categories: ");
    scanf("%d", &n);
    printf("Enter the observed frequencies:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%lf", &observed[i]);
    }
    printf("Enter the expected frequencies:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%lf", &expected[i]);
```

```

    }
    for (i = 0; i < n; i++)
    {
        chi_square += pow(observed[i] - expected[i], 2) / expected[i];
    }
    printf("Chi-Square Statistic: %.2lf\n", chi_square);
    return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users
\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 4.c -
o 4 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab
2\"4
Enter the number of categories: 5
Enter the observed frequencies:
10 20 30 40 50
Enter the expected frequencies:
30 30 30 30 30
Chi-Square Statistic: 33.33

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

6. Write a C program to implement Gap Test.

(i). Program:

```

#include <stdio.h>

int main() {
    int n, i, key;
    int arr[100];
    printf("Enter number of elements in sequence: ");
    scanf("%d", &n);
    printf("Enter the sequence (0-9 numbers):\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the digit to test (0-9): ");
}

```

```

scanf("%d", &key);
int last_index = -1;
int gap_count = 0;
printf("\nGaps between occurrences of %d:\n", key);
for (i = 0; i < n; i++) {
    if (arr[i] == key) {
        if (last_index != -1) {
            printf("Gap %d: %d\n", gap_count + 1, i - last_index - 1);
            gap_count++;
        }
        last_index = i;
    }
}
if (gap_count == 0) {
    printf("Not enough occurrences of %d to compute gaps.\n", key);
}
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users
\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 5.c -
o 5 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab
2\"5
Enter number of elements in sequence: 10
Enter the sequence (0-9 numbers):
2 5 3 2 7 2 9 2 4 2
Enter the digit to test (0-9): 2

Gaps between occurrences of 2:
Gap 1: 2
Gap 2: 1
Gap 3: 1
Gap 4: 1

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

7. Write a C program to implement Auto Correlation Test.

(i). Program:

```
#include <stdio.h>

#define MAX 100

int main()
{
    int n, i, k;
    double data[MAX], mean = 0.0;
    double numerator = 0.0, denominator = 0.0;
    printf("Enter number of data points: ");
    scanf("%d", &n);
    printf("Enter the data values:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%lf", &data[i]);
        mean += data[i];
    }
    mean /= n;
    printf("Enter lag (k): ");
    scanf("%d", &k);
    if (k >= n || k < 0)
    {
        printf("Invalid lag.\n");
        return 1;
    }
    for (i = 0; i < n - k; i++)
    {
        numerator += (data[i] - mean) * (data[i + k] - mean);
    }
    for (i = 0; i < n; i++)
    {
```

```

    denominator += (data[i] - mean) * (data[i] - mean);
}
double autocorr = numerator / denominator;
printf("Autocorrelation at lag %d = %lf\n", k, autocorr);
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 6.c -o 6 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\"6
Enter number of data points: 10
Enter the data values:
1 2 3 4 5 6 7 8 9 10
Enter lag (k): 1
Autocorrelation at lag 1 = 0.700000

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

8. Write a C program to implement Poker Test (For Three digit and Four Digit).

(i). Program:

```

#include <stdio.h>
#include <string.h>
void analyze(int num[], int n, int *counts)
{
    int freq[10] = {0};
    int pair = 0, triple = 0, four = 0;
    for (int i = 0; i < n; i++)
    {
        if (num[i] >= 0 && num[i] <= 9)
        {
            freq[num[i]]++;
        }
    }
    for (int i = 0; i < 10; i++)
    {
        if (freq[i] == 4) four++;
    }
}

```

```

        else if (freq[i] == 3) triple++;
        else if (freq[i] == 2) pair++;
    }
    if (four) counts[4]++;
    else if (triple) counts[3]++;
    else if (pair == 2) counts[2]++;
    else if (pair == 1) counts[1]++;
    else counts[0]++;
}
int main()
{
    int n, digits;
    int counts[5] = {0}; // all_diff, one_pair, two_pairs, three_kind, four_kind
    printf("Enter number of random numbers: ");
    scanf("%d", &n);
    printf("Enter number of digits (3 or 4): ");
    scanf("%d", &digits);
    if (digits != 3 && digits != 4)
    {
        printf("Invalid digit size. Only 3 or 4 allowed.\n");
        return 0;
    }
    int num[4];
    for (int i = 0; i < n; i++)
    {
        printf("\nEnter %d digits (0-9): ", digits);
        for (int j = 0; j < digits; j++)
            scanf("%d", &num[j]);
        analyze(num, digits, counts);
    }
    printf("\n--- Poker Test Results ---\n");

```

```

printf("All Different  : %d\n", counts[0]);
printf("One Pair      : %d\n", counts[1]);
if (digits == 4)
{
    printf("Two Pairs      : %d\n", counts[2]);
    printf("Three of a Kind : %d\n", counts[3]);
    printf("Four of a Kind : %d\n", counts[4]);
}
else
{
    printf("Three of a Kind : %d\n", counts[3]);
}
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 7.c -o 7 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\"7
Enter number of random numbers: 5
Enter number of digits (3 or 4): 3

Enter 3 digits (0-9): 0 1 2

Enter 3 digits (0-9): 3 4 5

Enter 3 digits (0-9): 6 7 8

Enter 3 digits (0-9): 1 2 3

Enter 3 digits (0-9): 4 5 6

--- Poker Test Results ---
All Different   : 5
One Pair       : 0
Three of a Kind : 0

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

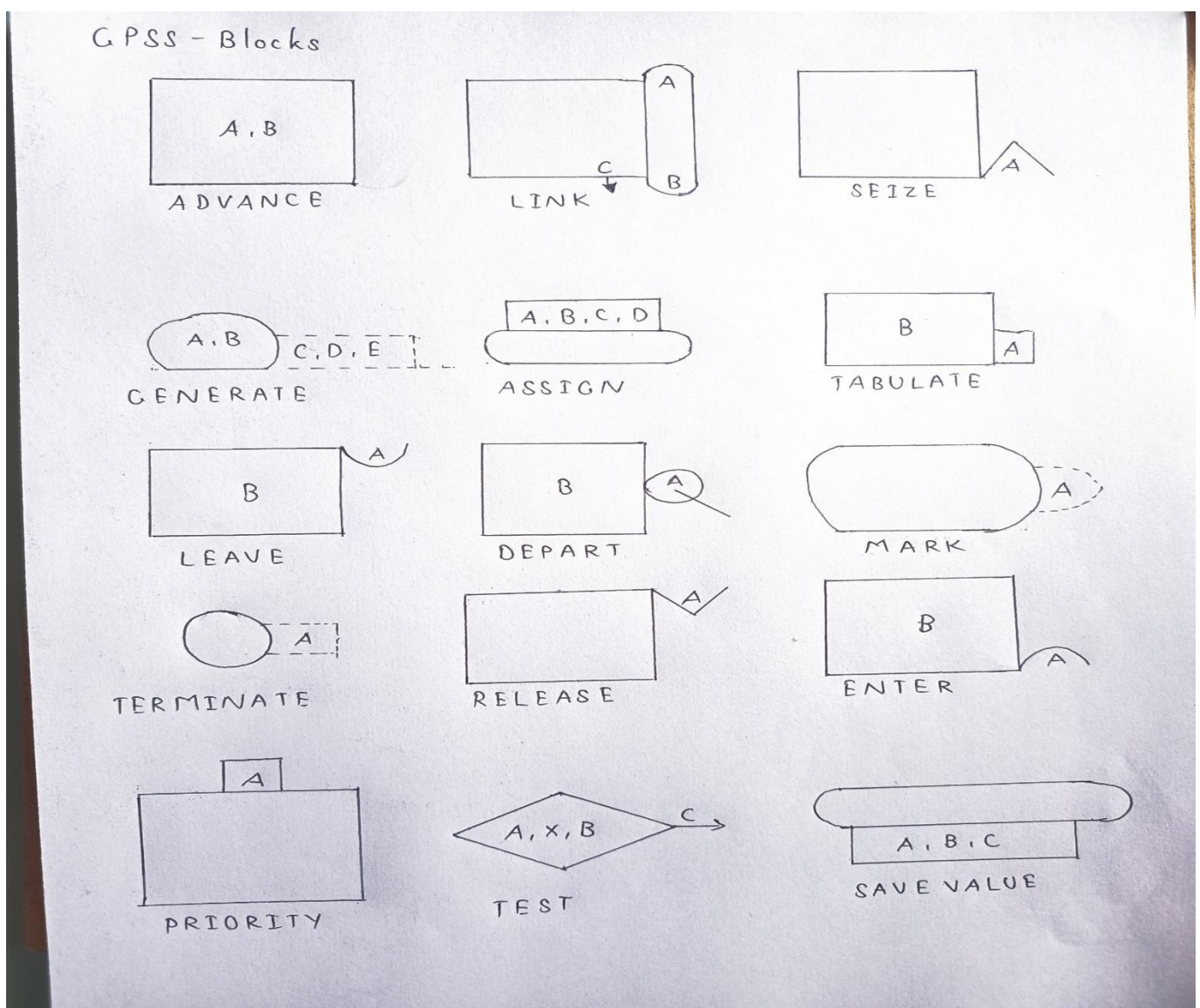
9. Explain GPSS blocks (major blocks that we use commonly).

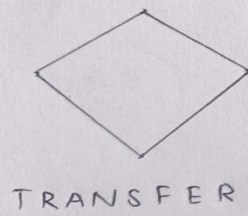
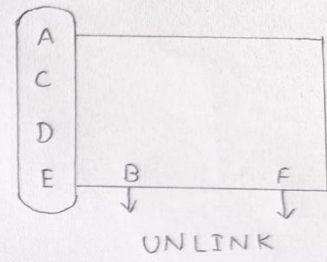
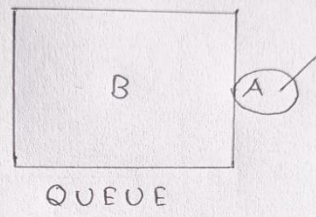
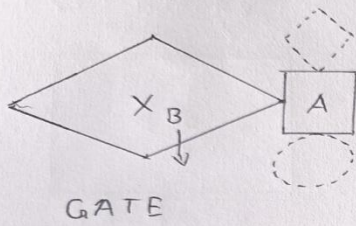
General Purpose Simulation System:-

GPSS is a discrete time simulation general-purpose programming language where a simulation clock advances in discrete steps. A system is modeled as transactions enter the system and are passed from one service to another. This is particularly well suited for problems such as factory. GPSS is less flexible than simulation languages such as Simula and SIMSCRIPT but it is easier to use and most popular.

GPSS was designed especially for analysts who were not necessary computer programmer. It is particularly suited for modeling traffic and queuing systems. A GPSS programmer does not write a program in some sense as SIMSCRIPT programmer does, instead he constructs blocks each performing a special simulation oriented function.

GPSS provides a set of 48 different blocks to choose from each of which can be used repeatedly. Each block has a name and specific task to perform. Moving through the system of blocks are entities called transactions are customer, messenger, machine etc.





10. Using GPSS, implement the following simulations.

10.1 Barber Shop.

(i). Program:

```
GENERATE 5,2
QUEUE BARBERQ
SEIZE BARBER
DEPART BARBERQ
ADVANCE 6,3
RELEASE BARBER
TERMINATE 1
```

(ii). Output:

GPSS World - [Barber.2.1 - REPORT]

File Edit Search View Command Window Help



GPSS World Simulation Report - Barber.2.1

Tuesday, May 05, 2026 22:03:51

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	14.655	7	1	0

NAME	VALUE
BARBER	10001.000
BARBERQ	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY
	1	GENERATE		2		0	0
	2	QUEUE		2		0	0
	3	SEIZE		2		1	0
	4	DEPART		1		0	0
	5	ADVANCE		1		0	0
	6	RELEASE		1		0	0
	7	TERMINATE		1		0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
BARBER	2	0.565	4.143	1	2	0	0	0	0

QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE. (-0)	RETRY
BARBERQ	1	1	2	1	0.148	1.088	2.175 0

CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
2	0	12.480	2	3	4		

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
3	0	16.522	3	0	1		

10.2 Mechanic on Service.

(i). Program:

```
GENERATE 15,5
QUEUE REPAIRQ
SEIZE MECHANIC
DEPART REPAIRQ
ADVANCE 12,4
RELEASE MECHANIC
TERMINATE 1
```

(ii). Output:

GPSS World - [Mechanic on service.2.1 - REPORT]

File Edit Search View Command Window Help

GPSS World Simulation Report - Mechanic on service.2.1

Thursday, May 14, 2026 22:42:46

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	47.909	7	1	0

NAME	VALUE
MECHANIC	10001.000
REPAIRQ	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY
	1	GENERATE		2		0	0
	2	QUEUE		2		0	0
	3	SEIZE		2		0	0
	4	DEPART		2		0	0
	5	ADVANCE		2		0	0
	6	RELEASE		2		0	0
	7	TERMINATE		2		0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
MECHANIC	2	0.558	13.378	1	0	0	0	0	0

QUEUE	MAX CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
REPAIRQ	1	0	2	2	0.000	0.000	0

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	3	0	48.806	3	0	1		

10.3 Manufacturing Shop.

(i). Program:

```
GENERATE 10,3
QUEUE MILL_Q
SEIZE MILL
DEPART MILL_Q
ADVANCE 8,2
RELEASE MILL
QUEUE DRILL_Q
SEIZE DRILL
DEPART DRILL_Q
ADVANCE 6,2
RELEASE DRILL
QUEUE INSPECT_Q
SEIZE INSPECTOR
DEPART INSPECT_Q
ADVANCE 4,1
RELEASE INSPECTOR
TERMINATE 1
```

☐

PFC	ION	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
4	0		58.239	4	10	11		
5	0		60.732	5	5	6		
6	0		62.503	6	0	1		

10.4 Restaurant Service.

(i). Program:

```
GENERATE 5,2
QUEUE WAIT_SEAT
SEIZE SEAT
DEPART WAIT_SEAT
ADVANCE 15,5
RELEASE SEAT
QUEUE WAIT_PAY
SEIZE CASHIER
DEPART WAIT_PAY
ADVANCE 2,1
RELEASE CASHIER
TERMINATE 1
```

(ii). Output:

GPSS World - [Restaurant Service.4.1 - REPORT]

File Edit Search View Command Window Help

GPSS World Simulation Report - Restaurant Service.4.1

Thursday, May 14, 2026 23:01:35

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	71.159	12	2	0

NAME	VALUE
CASHIER	10003.000
SEAT	10001.000
WAIT_PAY	10002.000
WAIT_SEAT	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY
	1	GENERATE	13		0		0
	2	QUEUE	13		8		0
	3	SEIZE	5		0		0
	4	DEPART	5		0		0
	5	ADVANCE	5		1		0
	6	RELEASE	4		0		0
	7	QUEUE	4		0		0
	8	SEIZE	4		0		0
	9	DEPART	4		0		0
	10	ADVANCE	4		0		0
	11	RELEASE	4		0		0
	12	TERMINATE	4		0		0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
SEAT	5	0.910	12.958	1	5	0	0	0	8
CASHIER	4	0.131	2.332	1	0	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY
WAIT_SEAT	9	8	13	1	4.185	22.910	24.819	0
WAIT_PAY	1	0	4	4	0.000	0.000	0.000	0

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
14	0		72.029	14	0	1		
5	0		84.340	5	5	6		

10.5 Ticketing System.

(i). Program:

```
GENERATE 4,2
QUEUE Q_TICKET
SEIZE TICKET_WIN
DEPART Q_TICKET
ADVANCE 5,2
RELEASE TICKET_WIN
TERMINATE 1
```

(ii). Output:

GPSS World - [Ticketing.3.1 - REPORT]

File Edit Search View Command Window Help

GPSS World Simulation Report - Ticketing.3.1

Thursday, May 14, 2026 23:07:21

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	35.434	7	1	0

NAME	VALUE
Q_TICKET	10000.000
TICKET_WIN	10001.000

LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY
	1	GENERATE		9		0	0
	2	QUEUE		9		3	0
	3	SEIZE		6		1	0
	4	DEPART		5		0	0
	5	ADVANCE		5		0	0
	6	RELEASE		5		0	0
	7	TERMINATE		5		0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
TICKET_WIN	6	0.848	5.011	1	6	0	0	0	3

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
Q_TICKET	4	4	9	1	1.441	5.675	6.384	0

CEC	XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	6	0	21.324	6	3	4		

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	10	0	38.570	10	0	1		

11. Write a C program to simulate games of dice that generate discrete random variate, using random number generation.

(i). Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#define MAX_SUM 12

int rollDie() {
    return 1 + (int)(6.0 * rand() / (RAND_MAX + 1.0));
}

int rollTwoDice() {
    return rollDie() + rollDie();
}

void printFrequency(int freq[], int start, int end) {
    printf("\n=== Frequency Distribution ===\n");
    printf("Outcome   Frequency   Probability\n");
    printf("-----\n");
    int total = 0;
    for (int i = start; i <= end; i++) {
        total += freq[i];
    }
    for (int i = start; i <= end; i++) {
        double prob = (total > 0) ? (freq[i] * 100.0 / total) : 0.0;
        printf("%3d %12d %12.2f%%\n", i, freq[i], prob);
    }
}

int main() {
    srand((unsigned int)time(NULL));
    int choice, numRolls;
    printf("=== Dice Simulation ===\n");
```

```
printf("1. Single Die\n");
printf("2. Two Dice\n");
printf("Choice (1-2): ");
if (scanf("%d", &choice) != 1) {
    printf("Invalid input!\n");
    return 1;
}
printf("Number of rolls: ");
if (scanf("%d", &numRolls) != 1) {
    printf("Invalid input!\n");
    return 1;
}
if (numRolls <= 0) {
    printf("Invalid number of rolls!\n");
    return 1;
}
int *freq = calloc(MAX_SUM + 1, sizeof(int));
if (!freq) {
    printf("Memory allocation failed!\n");
    return 1;
}
int start = 0, end = 0;
switch (choice) {
    case 1:
        start = 1;
        end = 6;
        for (int i = 0; i < numRolls; i++) {
            freq[rollDie()]++;
        }
        break;
    case 2:
```

```

        start = 2;
        end = 12;
        for (int i = 0; i < numRolls; i++) {
            freq[rollTwoDice()]++;
        }
        break;
default:
    printf("Invalid choice!\n");
    free(freq);
    return 1;
}

printFrequency(freq, start, end);
double mean = 0.0;
for (int i = start; i <= end; i++) {
    mean += i * freq[i];
}
mean /= numRolls;
printf("\nTheoretical Mean: %.1f\n", (choice == 1) ? 3.5 : 7.0);
printf("Observed Mean   : %.3f\n", mean);
free(freq);
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\" && gcc 8.c -o 8 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2\"8
=== Dice Simulation ===
1. Single Die
2. Two Dice
Choice (1-2): 1
Number of rolls: 50

=== Frequency Distribution ===
Outcome   Frequency   Probability
-----
1         9         18.00%
2         5         10.00%
3         7         14.00%
4         9         18.00%
5        11         22.00%
6         9         18.00%

Theoretical Mean: 3.5
Observed Mean   : 3.700

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 2>

```

12. Write a c program to perform Monte Carlo simulations simulate games of dice by generating discrete random variates.

(i). Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int i, n;
    int count[7] = {0};
    int roll;
    double average = 0;
    printf("Enter number of simulations: ");
    scanf("%d", &n);
    srand(time(0));
    for(i = 0; i < n; i++) {
        roll = rand() % 6 + 1;
        count[roll]++;
        average += roll;
    }
    printf("\nResults after %d simulations:\n", n);
    for(i = 1; i <= 6; i++) {
        printf("Face %d occurred %d times (Probability = %.4f)\n",
            i, count[i], (double)count[i]/n);
    }
    printf("\nAverage value = %.4f\n", average/n);
    return 0;
}
```

(ii). Output:

```
C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5\" && gcc 1.c -o 1 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5\"1
Enter number of simulations: 100

Results after 100 simulations:
Face 1 occurred 12 times (Probability = 0.1200)
Face 2 occurred 19 times (Probability = 0.1900)
Face 3 occurred 18 times (Probability = 0.1800)
Face 4 occurred 16 times (Probability = 0.1600)
Face 5 occurred 15 times (Probability = 0.1500)
Face 6 occurred 20 times (Probability = 0.2000)

Average value = 3.6300

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5>
```

13. Write a c program to perform Monte Carlo Simulation to estimate the circumference of a circle.

(i). Program:

```
#define _USE_MATH_DEFINES
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <math.h>

int main() {
    int n, i, count = 0;
    double x, y, r, radius, thickness;
    double circumference;
    printf("Enter number of random points: ");
    scanf("%d", &n);
    printf("Enter radius of circle: ");
    scanf("%lf", &radius);
    printf("Enter thickness (small value like 0.01): ");
    scanf("%lf", &thickness);
    if (n <= 0 || radius <= 0 || thickness <= 0) {
        printf("Invalid input.\n");
        return 1;
    }
}
```

```

srand((unsigned int)time(NULL));
for (i = 0; i < n; i++) {
    // Generate random point in square [-r, r]
    x = ((double)rand() / RAND_MAX) * 2 * radius - radius;
    y = ((double)rand() / RAND_MAX) * 2 * radius - radius;
    r = sqrt(x*x + y*y);
    if (r >= radius - thickness && r <= radius)
        count++;
}
double square_area = 4 * radius * radius;
circumference = (double)count / n * square_area / thickness;
printf("\nEstimated Circumference = %lf\n", circumference);
printf("Actual Circumference    = %lf\n", 2 * M_PI * radius);
return 0;
}

```

(ii). Output:

```

C:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling>cd "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5\" && gcc 2.c -o 2 && "c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5"2
Enter number of random points: 1000
Enter radius of circle: 5
Enter thickness (small value like 0.01): 0.01

Estimated Circumference = 20.000000
Actual Circumference    = 31.415927

c:\Users\V I C T U S\OneDrive\Documents\Prajwal\5th Semester\Lab Report For Simulation and Modeling\Lab 5>

```

14. Monte Carlo Simulation to Estimation the Circumference of a Circle using MS-Excel Function.

X	Y	$X^2 + Y^2$	Inside Circle
0.09058924704	0.6450278107	0.4242672883	1
0.9351299524	0.1593153135	0.899849397	1
0.8575034571	0.9515215923	1.64070552	0
0.5087970741	0.6517813616	0.683693406	1
0.4962353813	0.7772743912	0.8504050329	1
0.778900833	0.5502447937	0.9094558407	1
0.6740273683	0.2415883079	0.5126778038	1
0.4848515678	0.1569654667	0.2597192005	1
0.03665675991	0.0632889975	0.005349215251	1
0.2329545615	0.3950890167	0.2103631588	1
0.165097189	0.8121777831	0.6868898332	1
0.9186462943	0.2075333944	0.8869811239	1
0.4268118312	0.267683545	0.2538228195	1
0.5619109333	0.612576817	0.6909942536	1
0.9209961611	0.6058157369	1.215246636	0
0.236259705	0.6504338968	0.4788829023	1
0.2383767162	0.2880654588	0.1398051674	1
0.009041477361	0.9900068832	0.9801953771	1
0.3866552022	0.03617436992	0.1508108304	1
0.04463565056	0.8424907015	0.7117829235	1
0.2663052331	0.7901994433	0.6953336373	1
0.6165192703	0.3269108807	0.4869667346	1
0.2652560241	0.7876767804	0.6907954686	1
0.6716806503	0.03343273103	0.4522726435	1
0.325226363	0.9206887207	0.9534399076	1
0.5618784932	0.1679622928	0.3439187729	1
0.6204247228	0.955148169	1.297234861	0
0.4494535192	0.9738663177	1.150424071	0
0.3874524165	0.4906387707	0.3908457783	1
0.8875451875	0.704385244	1.283895032	0
0.5131056755	0.2358247374	0.318890741	1
0.9205318359	0.05644775392	0.8505652099	1
0.9870673344	0.1380245602	0.9933527019	1
0.3333085272	0.9201877731	0.9578401121	1
0.9750260541	0.5784137852	1.285238313	0
0.05877213977	0.498709517	0.2521653467	1
0.9835089587	0.4969900722	1.214289004	0
0.2208598507	0.2128642345	0.09409025599	1

Output:-

F	G
3.265306122	6.530612245